

Step 1: Frame Assembly for REC S550 Hexacopter

Needed Parts:



(Fig. 1)

(Additional parts include a soldering iron and solder, safety glasses, allen wrench set or screwdriver with multiple hex heads, zip ties)

- To begin find your 6 arms and the top plate (top plate has the triangle cut outs with “nobs” on each of the 6 squared off sides to connect to the arms as seen in Figure 2
- They will be screwed in using the screws labeled M2.5x6 (as seen in figure 3)



(fig 2)

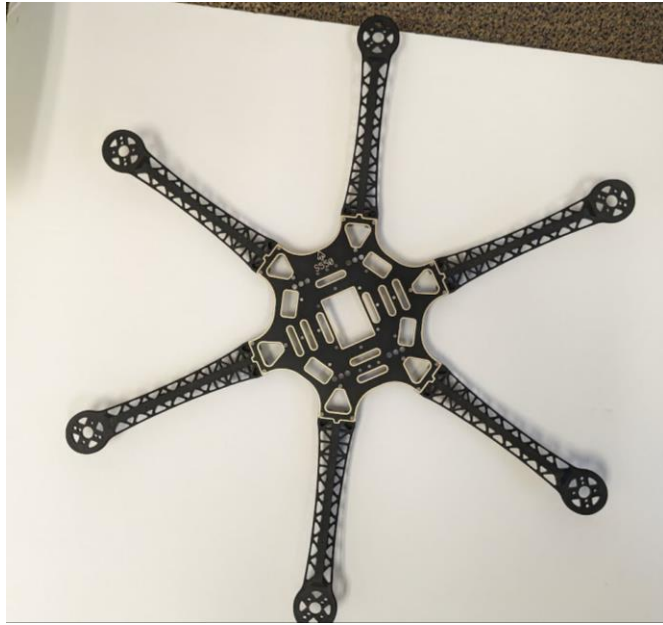


(fig 3)

- Repeat this step for all 6 arms as seen in figures 4 and 5



(Fig. 4)



(Fig. 5)

- Next Assemble the legs of the drone, taking the two red and black padded “feet” and the two carbon fiber tube legs. Loosen the 3 screws on each of the “feet” (fig 6) and slide the legs in, retightening the screws. Make sure the square on top is lined up perpendicular to the legs as seen in fig 7



(Fig. 6)



(Fig. 7)

- Next attach the legs to the bottom side of the bottom plate (not the plate with the 6 arms, the top side of this plate has + and - signs). You will do so by using the black screws shown in figure 8, and screwing the mounting bracket of the legs from the top as seen in figure 9, 10, and 11. Repeat with the other leg



(Fig. 8)



(Fig. 9)



(Fig 10)



(Fig. 11)

- Finally Attach the Battery and camera mounts. Flip the bottom plate over and attach 4 ring mounts (Shown in Figure 12) to the bottom of the bottom plate using more of the screws labeled M2.5x6 screwed in from the top as seen in Figure 13, it should look like Figure 14 when finished. Additionally squeeze the rubber shock absorbers into the rings as seen in Figure 15.



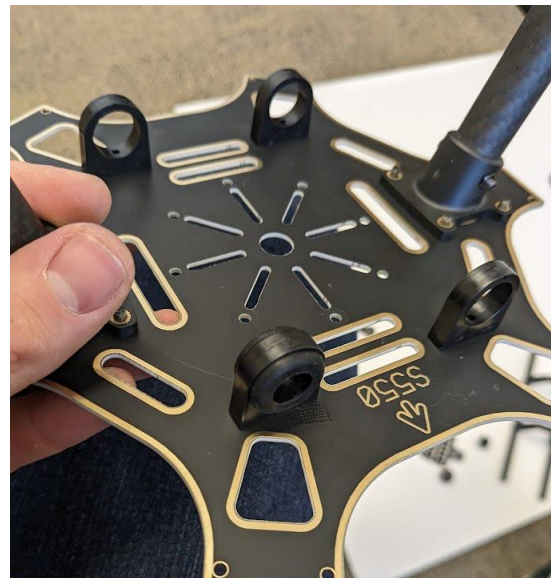
(Fig. 12)



(Fig. 13)



(Fig. 14)



(Fig. 15)

- Next assemble the battery mounts with the parts shown in Figure 16. Snap the two brackets on the two black tubes. Using the screws provided in the small bag, screw in the battery mount (The roughly I shaped rectangular piece) onto the two brackets (As seen in figure 17). This may take some sliding/adjusting of the brackets



(Fig. 16)



(Fig. 17)

- Take the two remaining ring mounts, and screw them into the camera mount (As seen in Figure 18). Additionally push the two rubber shock absorbers into the ring mounts as done previously (See figure 19)



(Fig. 18)

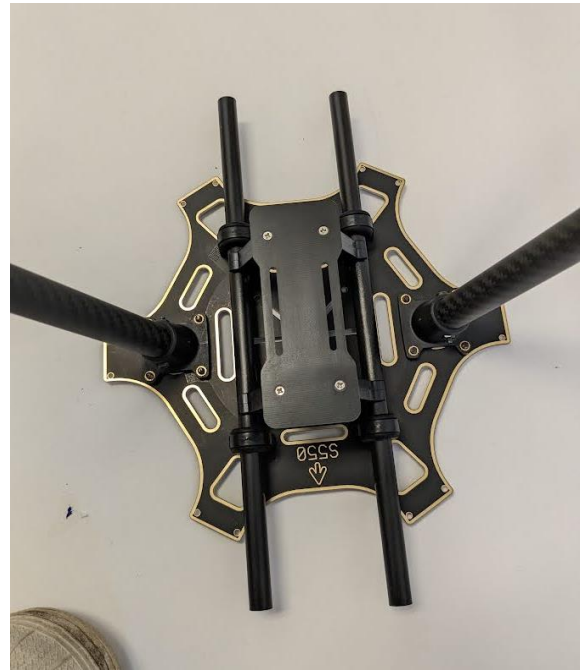


(Fig. 19)

- Slide the two rods out of the battery mount. Slide them halfway through the ring mounts on the bottom plate (see Figure 20). Once they are halfway in, slide the battery mount on the rods, in between the 4 ring mounts of the bottom plate. Slide the rods until they fully extend out the other side (See fig. 21)

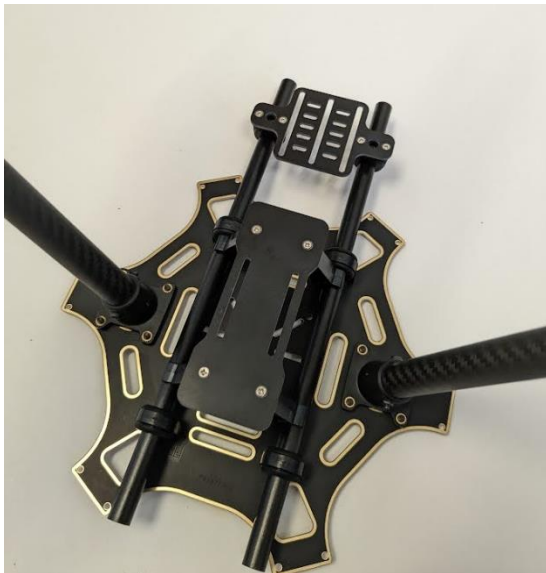


(Figure 20)



(Figure 21)

Continue to push the rods through until they extend more toward the front (where the arrow on the bottom plate points). Slide the camera mount on the rods as seen in Figure 22



(Fig 22)

- Finally, attach the motors to the ends of the arms. Each motor comes with small black screws.

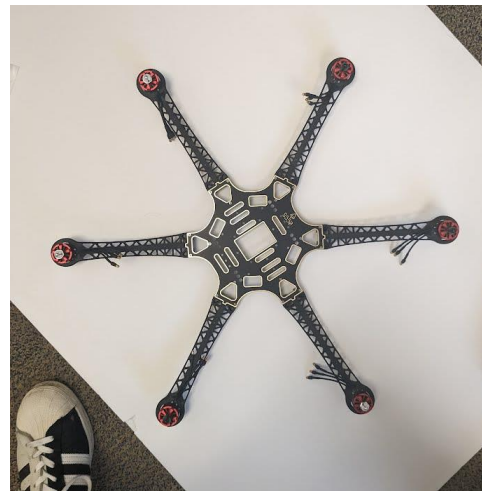


(Fig 23)

- Use these to screw in the motor from the bottom on each arm (so that the motors sit on top of the arm facing up). Screw all 4 screws. Alternate between a black capped motor and a silver capped motor given the below instructions on configuration (see Figures 24 and 25)



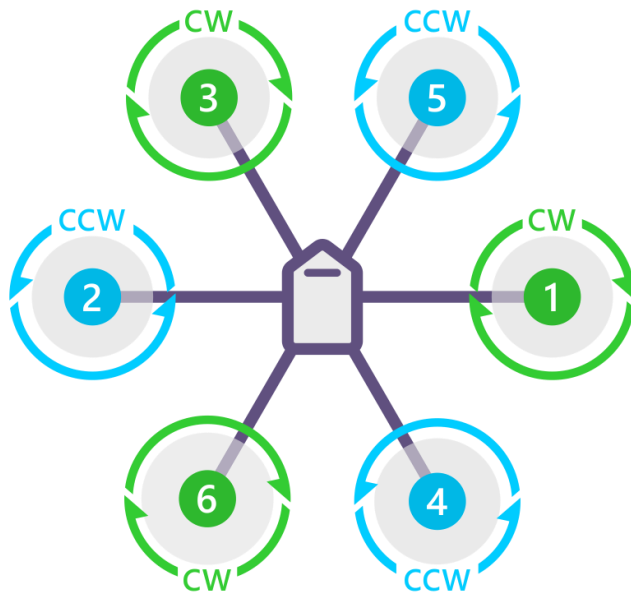
(Fig. 24)



(Fig. 25)

Note About Motor configuration:

For Drones to produce lift, Their motors spin in two directions, Clockwise (CW) and Counter Clockwise (CCW). Additionally these motors need to be placed in a specific order to produce lift, and to properly configure the flight controllers and Electronic Speed Controls (ESCs). Below is a picture of how a Hexcopter should be assembled (Figure 26)

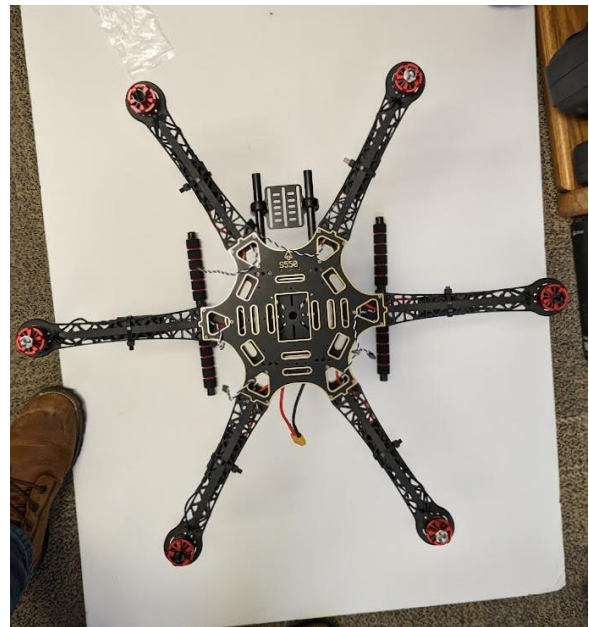


(Fig. 26)

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Furthermore, each motor has their own thread direction (The direction in which the propeller nut screws on). In this case, the silver caps screw on CW and the black caps screw on CCW. The direction in which the prop nuts screw on should be OPPOSITE to the direction the propellers will be spinning. This way, the nuts will be “sucked down” and automatically tighten as the propellers spin, rather than being loosened during flight.

So, the silver caps should be placed in the CCW (positions 2, 4, and 5) and the black caps should be placed in the CW positions (positions 1, 3 and 6) as seen in Figure 27



(Fig. 27)

2. Soldering

Needed Parts:



- Next, you will be soldering the esc controllers and the battery connection to the bottom plate. These will be soldered onto the gold ovals next to the positive and negative symbols (red wire -> Positive, black wire-> negative).

Note: Take care and use Safety glasses. Ask for help if this is your first time soldering.

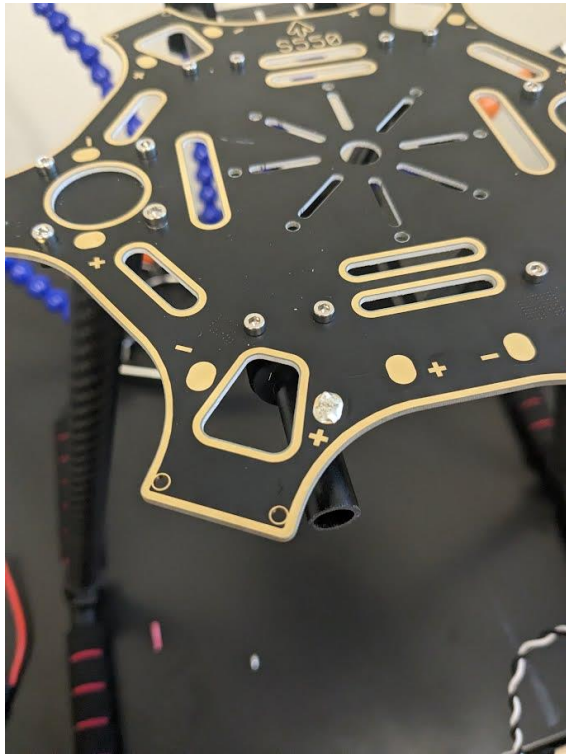
Here are two links to videos about soldering:

https://www.youtube.com/watch?v=Qps9woUGkvl&ab_channel=oneTesla

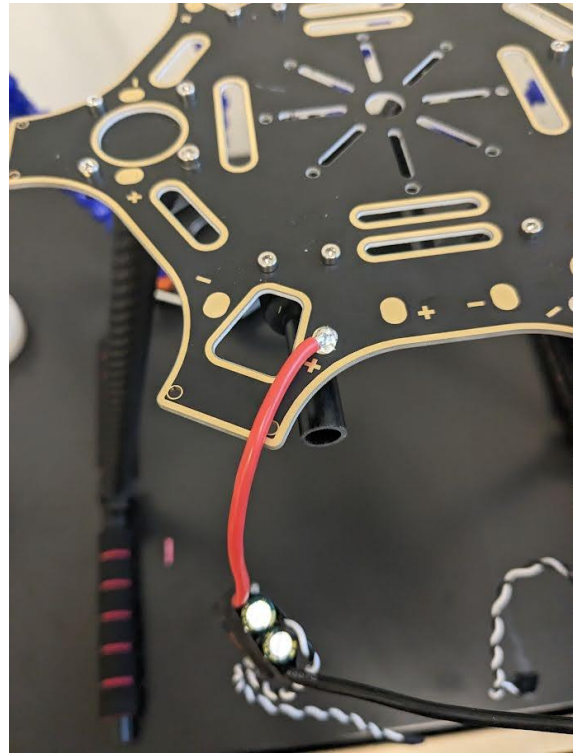
https://www.youtube.com/watch?v=6rmErwU5E-k&ab_channel=wermymy

^This Video goes more in depth with various soldering scenarios, at 9:17 is reviews the type of soldering done for this drone^

- Start by soldering in the ESCs. There will be 6 “terminals” in a hexagon pattern, the terminal for the battery connection will be located opposite to the arrows.
- As stated in the video, start by adding a large drop of solder to the golden oval (see Figure 26). Next heat up the “glob” of solder and insert the wire (red if positive, black if negative) (see figure 27). The wires come “pre-tinned” meaning they already have some solder making it easier to insert.
- Make sure the wires are facing away from the board (or towards the tips of the 6 points of the hexagonal shape)
- Note: Take care not to burn yourself or the wire as they may heat up during soldering.

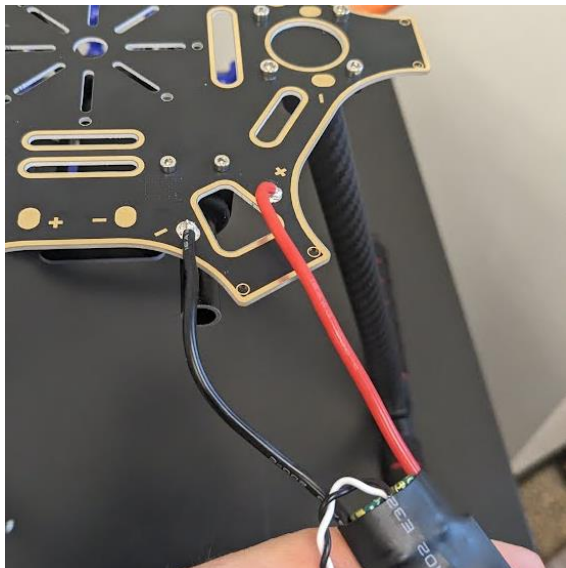


(Fig. 28)



(Fig. 29)

- Repeat this for all ESCs

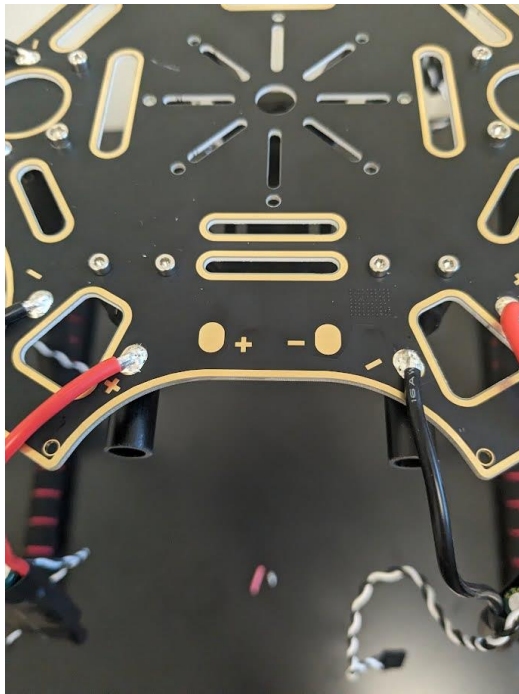


(Fig 30)

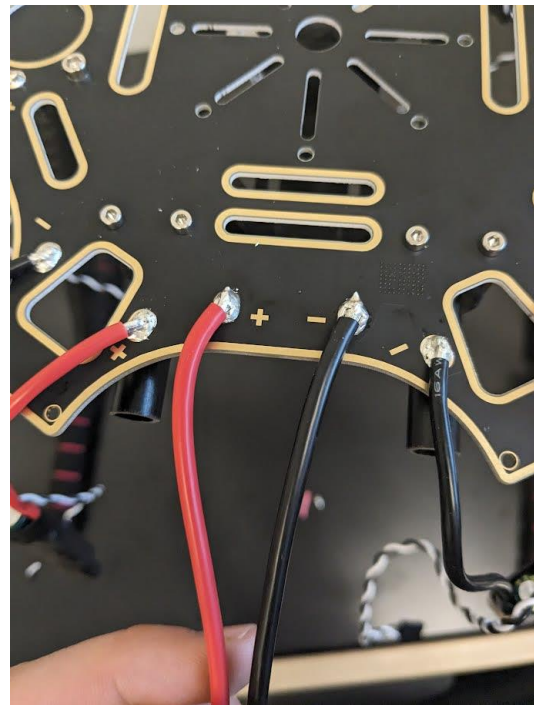


(Fig. 31)

- Finally, solder the battery connector on the two remaining terminals (see fig 32 and 33)



(Fig. 32)



(Fig. 33)

- Now you can attach the top and bottom plate. This will be done using the mounts on the bottom of the drone arms and the remaining holes on the bottom plate. Use the screws labeled M2.5x6 and screw in from the bottom plate to the drone arms as seen in figures 34 and 35



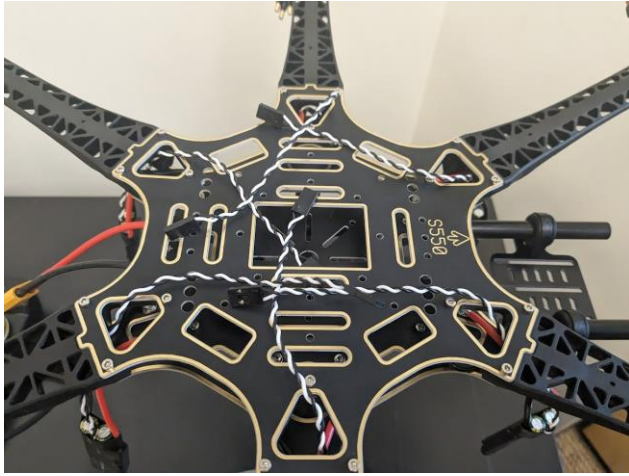
(Fig. 34)



(Fig. 35)

- Make sure both of the soldered wires go in between the two sides of the drone arm (see fig 35). You also will feed the smaller black and white cable with the 3 pronged

connector through this same gap and up through the triangle on the top plate (see fig. 36). This may take some fiddling. It may help to flip the drone over so the bottom plate is facing up

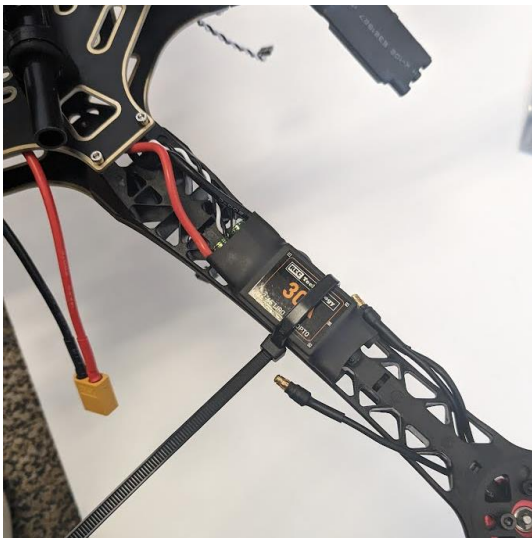


(Fig. 36)



(Fig. 37)

- Finally, you will zip tie the ESCs to the bottom of the drone arms. Do so about halfway down the arm, so the sticker (with the name and information of the ESC) is visible. Tighten each zip tie and cut off the excess.

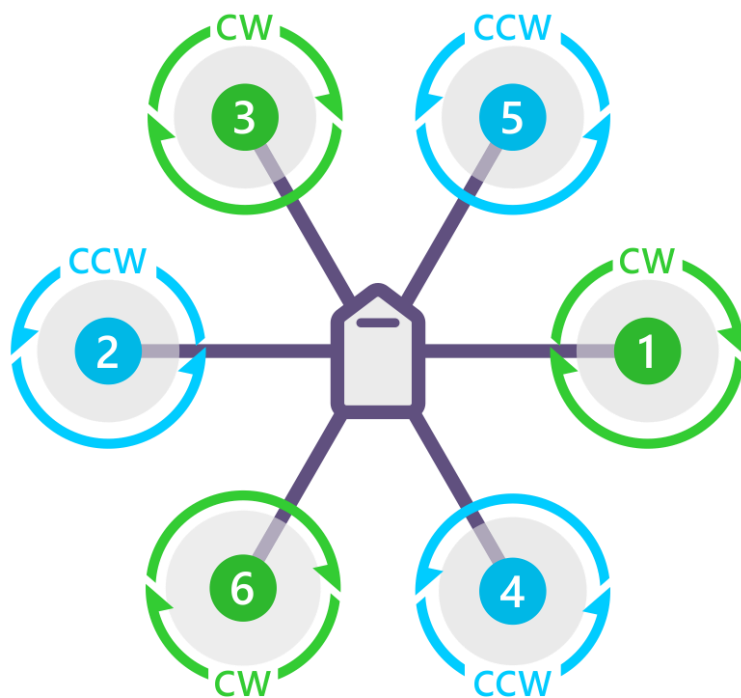


(Fig. 38)



(Fig. 39)

- It is important to be familiar with the orientation of the drone. The front of the drone has the arrow on the top and bottom plate pointed in that direction. It also has the camera mount. To help remember we will call the camera mount the “head” of the drone and the battery connection on the other side the “tail” to remember the orientation.
- Given the orientation, the diagram below (figure 40) states what number each motor is and what direction it should be spinning. As stated above, Silver caps should be in the positions 2,4, and 5 spinning CCW and the black capped motors should be in positions 1,3, and 6 spinning CW.



(Fig. 40)

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- Plug in the ESC. At this stage it is unclear in what direction they will spin based on connection, so plug the 3 wires leaving the motor into the 3 ports of the ESC in whatever order, but be consistent. Plug all the motors with black heads in one fashion, then for the silver ones, swap two wires. Be consistent, and as you test later, you can easily re-plug the motors so they turn the desired directions.
- Below are examples for the black and silver motors

Black capped Motors:

In this example the 3 wires were plugged directly into the ESC from left to right



(Fig. 41)



(Fig. 42)

Silver Capped Motors:

The configuration is the same as the black motors, accept the two rightmost wires were flipped



(Fig. 43)



(Fig. 44)

3. Flight Controller and GPS Configuration

Needed Parts:



Additional Parts: Cardboard box or piece of cardboard, micro USB to USB cord, computer with storage to download mission planner software.

- To begin calibration of the flight controller and GPS, download The Mission Planner software from the following link:

[Installing Mission Planner — Mission Planner documentation \(ardupilot.org\)](https://ardupilot.org/Documentation/missionplanner/InstallingMissionPlanner.html)

- The link also contains more information on how to download and use the software. Additionally, here are two videos with examples. However, their software is a slightly different version, they are still helpful and it can easily be applied to the latest software found in the link.

[\(1/5\) PixHawk Video Series - Simple initial setup, config and calibration - YouTube](https://www.youtube.com/watch?v=...)

(Skip to 11:45 for the tutorial on the software)

[How to Build and S550 Hexacopter with Pixhawk Flight Controller - YouTube](https://www.youtube.com/watch?v=...)

(Skip to 50:46 for software tutorial)

- [illegible]

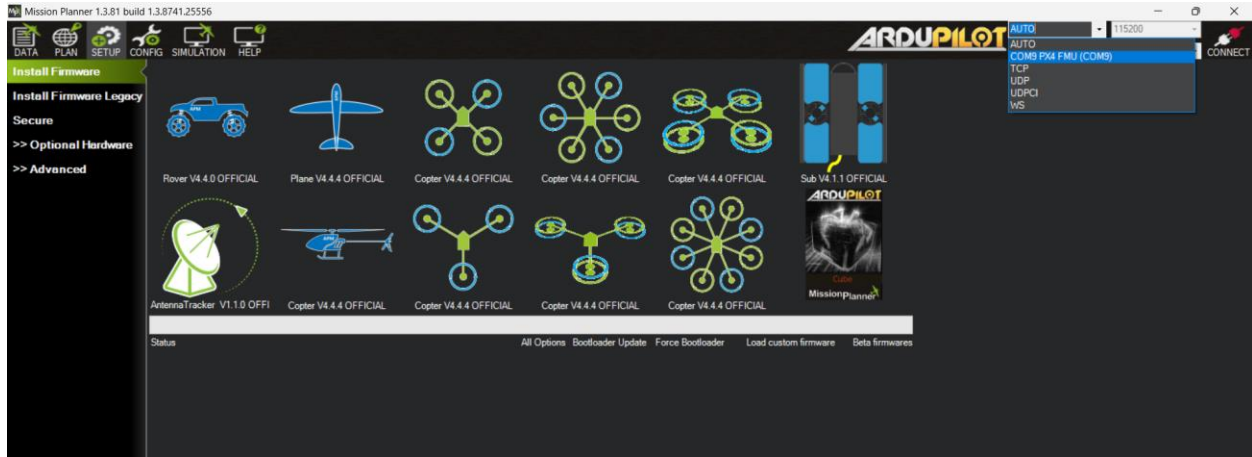
A black Pixhawk flight controller board is shown. It features a central microcontroller, various connectors for sensors and actuators, and a cooling fan. The board is labeled with 'pixhawk' and '2.4.8'. It has a USB port, a UART port, a CAN port, and a 12C port. It also has a power input and a power output. The board is connected to a black cable with a red, yellow, and green wire bundle.

- Using tape, or another non permanent method, attach the components to the box or piece of cardboard. Make sure the arrow on the antenna is pointed in the same direction as the arrow on the pixel hawk (This is necessary for proper calibration) (See Figure 47)



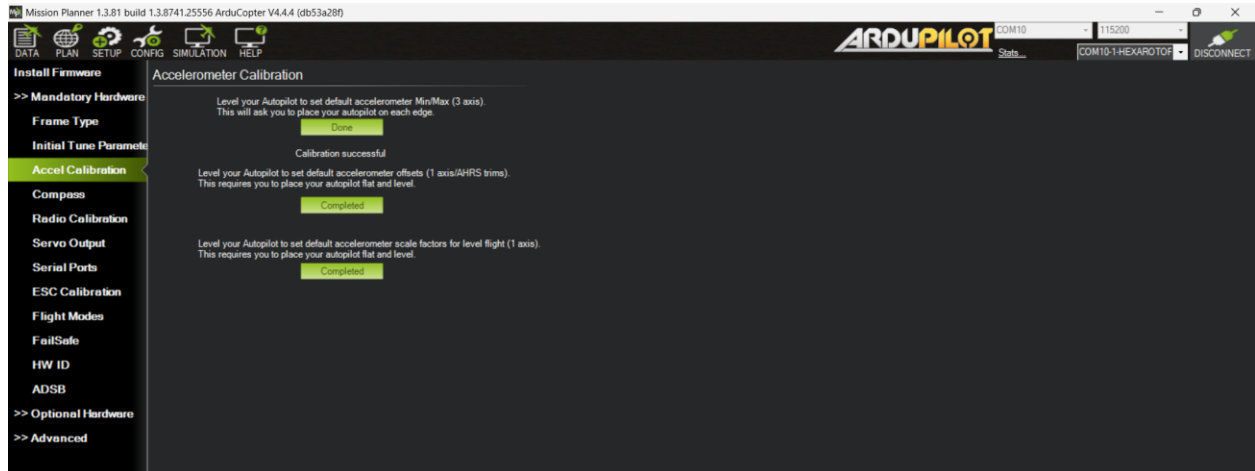
16

- Next, plug in the pixelhawk to the computer and open the Mission Planner software
- In the top right corner, use the dropdown selector to find the correct port that is connected to your pixel hawk (Figure 48)



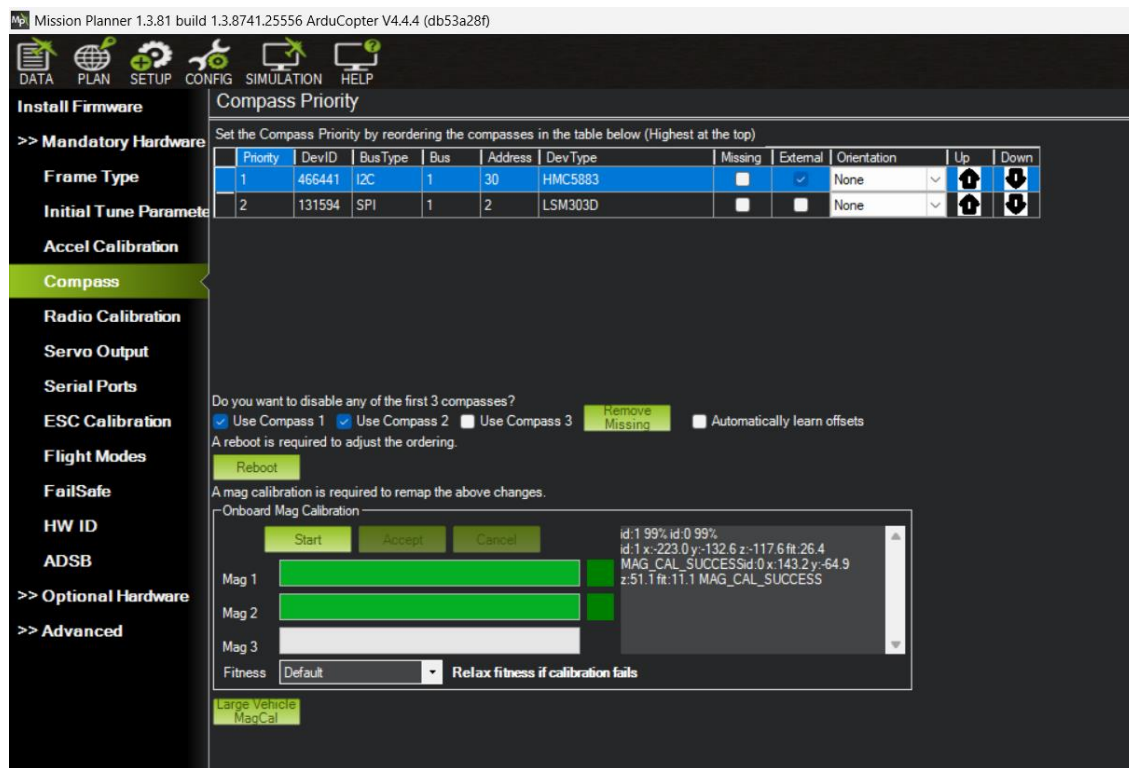
(Fig. 48)

- Go to install firmware and click on the icon representing the Hexacopter option. Download the firmware (You may need to additionally select the Pixelhawk 1 option)
- Follow the download instructions, it should ask you to disconnect and reconnect your flight controller to flash the software. Once this is done click the connect button in the top right corner.
- Next go to the setup tab. Here you will complete the frame type, accelerometer, and compass calibrations. First in frame type, select the basic hexacopter configuration
- In Accel Calibration, it will ask you to place the flight controller and attached GPS level, then on its front (nose down), on its side, etc. Complete each configuration until they all say completed as seen in figure 49



(Fig. 49)

- For compass configuration make sure it recognizes 2 compasses (there is an internal one inside the pixhawk and the second in the attached GPS) Make sure the one labeled I2C is shown as external and click to only use the first two compasses as seen in Figure 50.
- Click Start, and rotate the box with the controller and GPS in as many directions as you can (Slowly and making sure not to tangle or damage cords). The Pixhawk will beep and the two green bars will fill up slowly



(Fig. 50)

- Your Pixhawk GPS is now configured. It may be removed from the box. Later the receiver will need to be attached and configured, but this will be done once it is all assembled

4. Attaching Flight Controller and Calibration

Needed Parts:



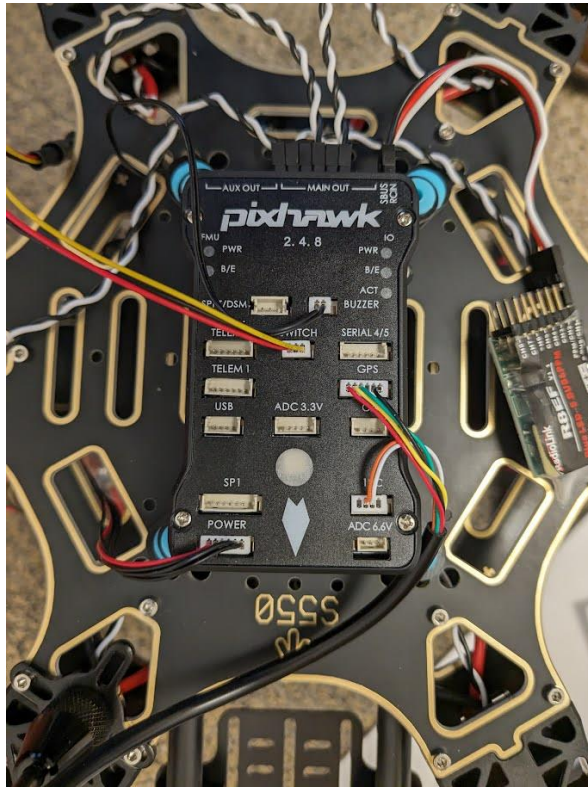


- First, assemble and attach the shock absorption plate to mount the flight controller (the black platforms with the blue rubber shocks) It may be easier to attach the bottom plate first (with the blue shocks inserted) then attach the top plate. In the examples it was assembled first then attached
- Using more of the provided M3 screws (not the ones labeled m2.5) and 4 M3 nuts, secure the plate using the 4 holes on the bottom plate of the absorber and the corresponding 4 holes on the top plate of the drone frame (see figure 51)

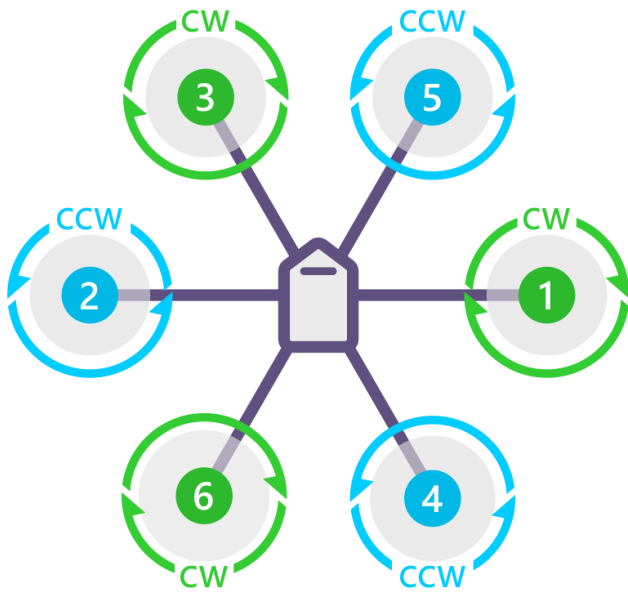


(Fig. 51)

- Next, use the 2 round double sided tape pads and attached them to the topside of the shock absorber. Remove the other sticker and attach the pixel hawk, with the arrow pointed to the front (head) of the drone (matching the arrow on the top plate) as shown in figure 52

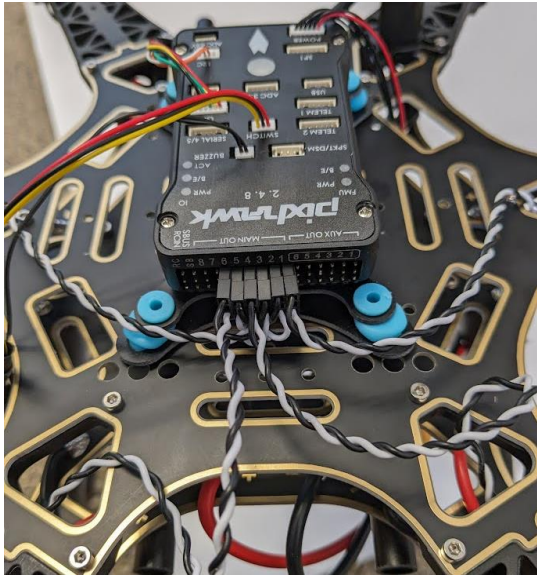


(Fig. 52)



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- Attached the ESCs. Plug them in with the black wire on top. Plug them in on the “main out” section of the Pixhawk. Plug the esc corresponding to motor 1 (as seen in the diagram to the left) into slot 1. Numbers 3 and 5 may be hard to connect (as they are furthest away) you might try running the wires in between the base and top plate, sliding the escs closer to the plate, or even untwisting the cables slightly so they can reach. Ports 1-6 should be used as seen in Figure 53



(Fig 53)

- Next assemble the GPS/antenna stand. This is done by first inserting the hinge into the bottom stand (or foot) and securing it using the provided silver screw (see Figure 54)
- Insert the rod with the flat gray end facing the hole of the hinge, use the small black screws to secure it (Figure 55)
- Place the top of the stand on the other end of the rod, secure it by facing the flat gray side towards the hole and tightening the small black screw (Figure 56)



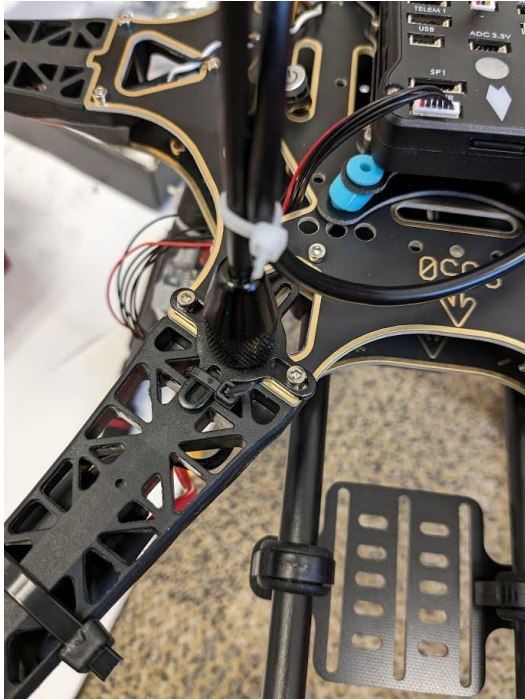
(Fig. 54)



(Fig. 55)



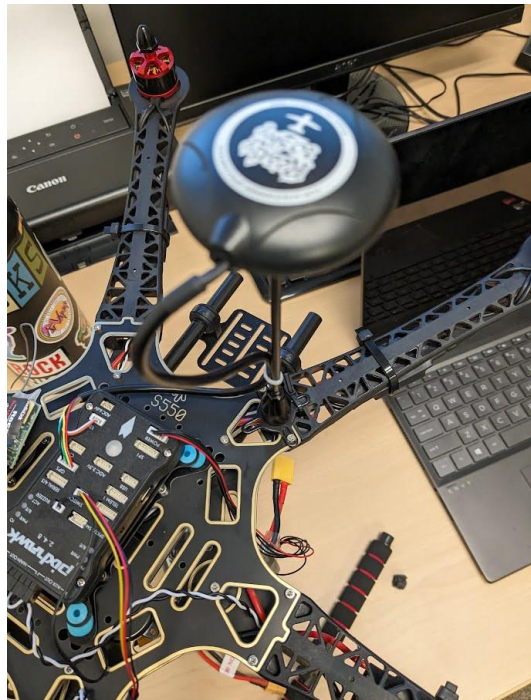
(Fig. 56)



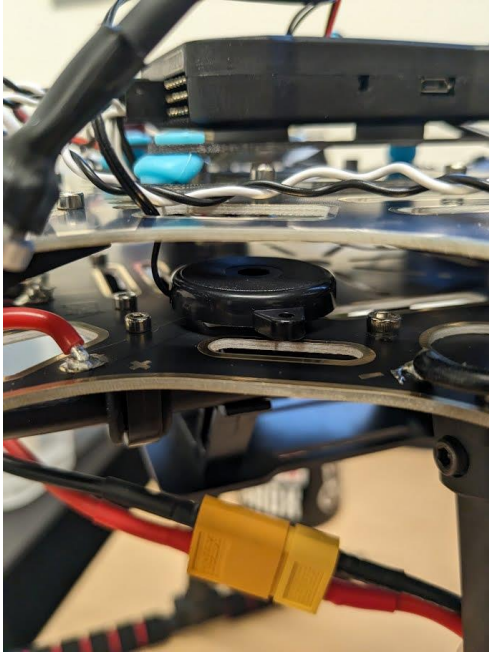
- To attach the antenna stand to the drone, unscrew two of the screws from the top plate securing the top right arm of the drone. Place the foot of the stand so two of the slits in the foot are above the screw holes where the screws were just removed, and replace the screws so they now are securing the foot of the antenna stand, top plate, and drone arm together (see figure 57). You may wish to add another small zip tie or two to secure the stand even further

(Fig. 57)

- Using the double sided tape circle included with the GPS, mount the GPS to the stop of the stand, ensuring the arrow is pointed in the same direction as the flight controller and top plate of the drone (See figure 58)
- Secure the GPS wire to the stand with a small Zip Tie



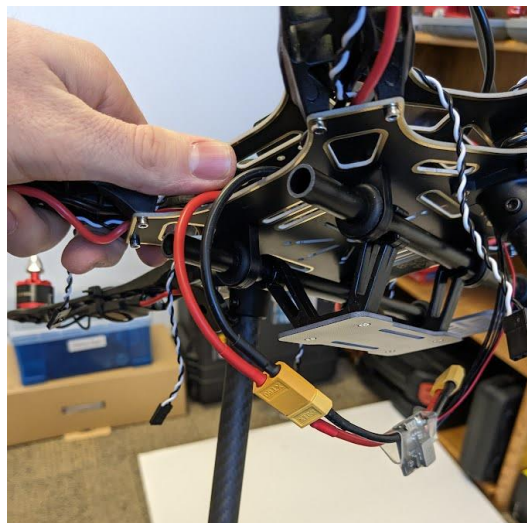
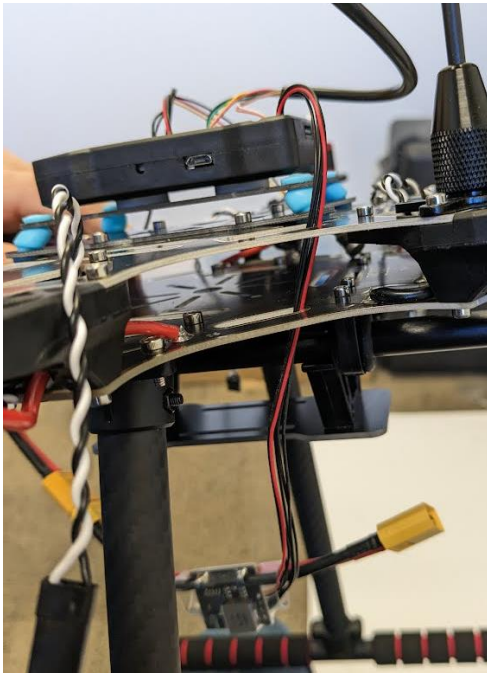
(Fig. 58)



- Secure the buzzer, in this example it was placed on the bottom plate and secured using double sided tape (See Figure 59). Make sure the wire is not stretched or wrapped around other wires when doing so.

(Fig. 59)

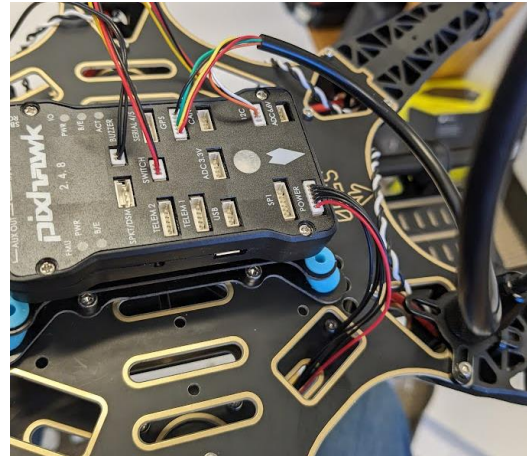
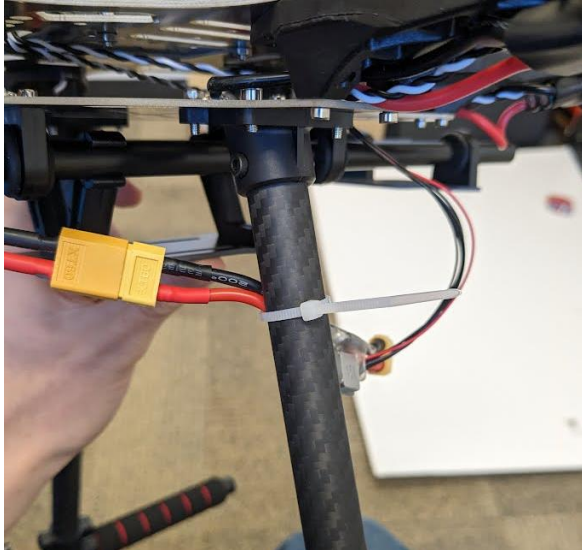
- Next attached the battery module. This is used to power the flight controller and allow the flight controller to monitor the status of the battery. Attach the end of the battery connection that was soldered onto the bottom plate. (Figure 60)
- Then run the power wire through both the bottom and top plates and connect it to the PixHawk flight controller (Figures 61 and 62)



(Fig. 60)

<-(Fig. 61)

(Fig. 62)->



- Zip tie the battery Module to the leg of the drone to secure it and not put strain on the wires

(Fig. 63)